

Assessment Result

Assessee Information

Assessment Number:	TPCP/1001366/2025/2	OPU/HCU:	GTS Centre
Staff Name:	Muhammad Aminuddin bin Ishak	Assessment Skill Group:	SKG16 : Process and Operations
Staff Number:	1001366	Assessment Discipline:	16.4 Plant Performance & Optimization
SG:	P04	Assessment Sub Discipline:	Advanced Process Control
Assessment Type:	Reassessment	Assessment SKG AOS:	N/A
Assessment Level:	Staff	TI&R Version:	3
		Assessment Date:	04/08/2025

Assessment Details

	Technology Class	Technology Inventory	Compulsory TI
	B1	Process Technology*	YES
	B2	Basic and Intermediate Regulatory Control*	YES
	B3	PID Tuning	NO
	B4	DCS Configuration	NO
	B5	Complex and Advanced Regulatory Control	NO
	B6	Plant Information Management System	NO
	K1	Scouting Study / Functional Design / Benefits Analysis*	YES
	K2	Plant Test and Modelling*	YES
	K3	Controller Build / Simulation*	YES
	K4	Implementation and Commissioning	NO

	K5	APC Integration with DCS	NO
	K6	APC Performance Monitoring*	YES
	K7	Regression (Statistical) Property Inferential (RPI)*	YES
	K8	First Principle Inference (FPI)	NO
	K9	Multi-Unit Optimisation	NO
	P1	Expert System	NO
	P2	Non-Linear MPC	NO
	E1	Parallel Coordinates	NO
	E2	Advanced Analytics for Optimization	NO

Technology Know How:

Competency Level:	1 - Awareness	2 - Knowledge	3 - Skill	4 - Advance	5 - Expert
TI&R Descriptor					
					

B1	Process Technology*
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Assessee Rating Description:*

I ahve conducted a few projects across few plants and different processes. For example, PCEPE APC Revamp 2012 -2023 (Hot and Cold side). Other than that, I have also did some works in Trans Thai Malaysia GPP BLCAR, MCOT Vessel Control Scheme Change, Erb West,Semarang BLCA, Ammonia PCFK OJT and Step Test,Butanol DCS Migration,PCLDPE APC Assessment and APC Revamp,PCMSB APC Implementation, ABF PEMS Implementation

Competency Level

Previous Score:	2	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Explained the PCEPE overall process from supply to tank and consumers. Able to explain the different unit objectives and the operating conditions within the units, incl the constraints of the units. Explained the furnace cracker in more detail and the Le Chatelier's principle. Explained the conversion and yield calculation and how the optimum conversion impacts the yield. Explained pushing higher conversion by increasing COT impacts selectivity. Explained the different elements, e.g. dilution steam, residence time, can impact conversion & selection in the cracker. Explained troubleshooting for PTLE for E-102A in F2 where the fouling caused by mix feed valve sticky; coil crack in F5 due to frequent decoking resulting in loss of Ni; saturated burners due to clogging of burners (polymer fr Deeth).
Amin meets Skill level.

B2	Basic and Intermediate Regulatory Control*
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Assessee Rating Description:*

Amin has completed tasks as per below:

PCESB

- 1)Control Scheme Change on Ethylene Fractionator(C402)
- 2) Control Scheme Change on Deethanizer Column(C401)
- 3)Control Scheme Change on Mixfeed Controller Combustion
- 4)Control Scheme Change on Hearth/Wall Burner
- 5) Control schemes change on Compander

PCMSB Plant 1

- 1) Equivalent Duty Control Implementation at Atmospehric Column(D302)
- 2) Split range control

PCMSB Plant 2

- 1) 3 inventory controls and 1 temperature controller (CASCADE)

Competency Level

Previous Score:	0	Self:	4	Assessment Target:	4	Assessor:	4	Gap:	0
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Assessor Remarks:*

Shared his experiences in PCESB & PCMSB, explained the control layers, instrumentation terms. concept of dynamic stability. Explained concept of cascade ctrl & its design requirements. Explained his ctrl chg for cascade in PCESB C401 - shared his DCS config work, explained how this cascade design overcomes issues in maintaining process stability of the column. Meets advanced level for cascade.

Shared snapshots for HIRA & MoC where he is involved in preparing the documents to mitigate the risks in deploying the ctrl change.

Explained override ctrl & design config. Explained his work in PCMSB & PCESB in troubleshooting problematic override ctrls & the modification or action required. Meets advanced level for override.

B2c: Can explain the function of the equipment. Explained compressor surge. Explained PCESB compander operation (4 companders in parallel). Explained the control change from single speed ctrl to 4 speed ctrl, can explain how the ctrls function during s/up & transition (plant load chg) which affects the speed ctrl. Meets Skill level.

B2f: Already explained during B1 and meets knowledge level.

B2g: Explained what parameters are monitored and how to interpret the data. Can explain what is monitored for regulatory ctrls and APC. Able to explain what the triggers are to carry out further investigation (uptime, noise) & whether to implement changes to the ctrl scheme. Can explain additional parameters that can be used to further enhance perf monitoring. Explained certain parameters like PV std dev on its own is not reliable and needs to be viewed together with other parameters, e.g. noise.

Explained split range ctrl, design reqt, issues with it. Shared the findings in PCLDPE, PCML where convert split ctrl to splitrage as split ctrl didnt work. Meets advanced level for split range.

Explained ratio ctrl, design reqt & shared experience in issues found & rectifying in PCESB hearth/wall ctrl. Not sure since only did 1 for ratio but can be said as meets skill.

Explained issues with noise to ctrl & how to overcome it. Explained issues if putting too strong a filter. Can explain the diff ways of doing filter depending on DCS (either filter time or implement a lag block). Shared his experience @C402 reflux flow, PCMSB duty ctrl. Meets advanced level for noise.

B2b (distillation): Explained distillation process, eqt, ctrl objectives, issues in distillation & how it impacts ctrl. Shared his work in PCEPE C401 & C402 ovhd ctrl, PCMSB D302 duty ctrl where he identified the need to modify these ctrls and carried out the implementation. Met Adv level for B2b.

B2d (combustion): Explained the combustion process, the key variables, explained abnormal operations for the PCESB cracker. Explained PCESB boiler ctrl in detail where discovered where can be optimised in terms of changing excess air. Shared test plan for PCESB boiler. Meets advanced for B2d.

B3	PID Tuning
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Assessee Rating Description:*	Has completed controller tuning and rectification as per below
	PCESB Base Layer Rectification
	TTM Base Layer Rectification
	Butanol PCOGD DCS Migration
	PCMSB APC Implementation
	PCLDPE APC Implementation

Competency Level

Previous Score:	0	Self:	4	Assessment Target:	4	Assessor:	4	Gap:	0
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Assessor Remarks:*	explained reason for PID tuning & impact of incorrect tuning. Explained how to obtain proc dyn, explained PID, explained diff tuning techniques. Explained tuning rules and why IMC lambda is preferred. Explained how to eval ctrlr performance. Explained the diff PID algo and how it impacts perf of the ctrlr. Shared his experience in tuning loops in projects he was involved in. Can discuss auto tuning advantages & disadvantages. Discussed issues with processes that shows non-linear behaviour, Explained auto tuning.
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B4	DCS Configuration
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Assessee Rating Description:*

Has experienced DCS configuration while doing tasks below:

PCESB

1)Control Scheme Change on Ethylene Fractionator(C402)

2) Control Scheme Change on Deethanizer Column(C401)

3)Control Scheme Change on Mixfeed Controller Combustion

4)Control Scheme Change on Hearth/Wall Burner

5) Control schemes change on Compander

PCMSB Plant 1

1) Equivalent Duty Control Implementation at Atmospehric Column(D302)

2) Split range control

PCMSB Plant 2

1) 3 inventory controls and 1 temperature controller (CASCADE)

Competency Level

Previous Score:	0	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Explained his configuration work in DCS for cascade control (PCESB), override (PCESB), split range (PCMSB), ratio (PCESB) & noise - met the requirements for Skill.

Explained functions of DCS, its architecture (general & PCESB's). Explained the different blocks avail in DCS (using examples fr his projects) and how he used them in carrying out ctrl scheme chgs. Also described blocks that can enhance ctrls. Also shared his experience in troubleshooting which requires ctrlIr config chgs

B5	Complex and Advanced Regulatory Control
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Assessee Rating Description:*

Amin has completed tasks as per below:

PCESB

1)Control Scheme Change on Ethylene Fractionator(C402)

2) Control Scheme Change on Deethanizer Column(C401)

3)Control Scheme Change on Mixfeed Controller Combustion

4)Control Scheme Change on Hearth/Wall Burner

5) Control schemes change on Compander

PCMSB Plant 1

1) Equivalent Duty Control Implementation at Atmospehric Column(D302)

2) Split range control

PCMSB Plant 2

1) 3 inventory controls and 1 temperature controller (CASCADE)

Competency Level

Previous Score:	0	Self:	3	Assessment Target:	3	Assessor:	0	Gap:	3
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Assessor Remarks:*

Not assessed

B6	Plant Information Management System
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Assessee Rating Description:*

Has conducted a few projects across few plants and different processes. Have utilized PI, IP21 and Exaquantum to do works at different plants
PCEPE APC Revamp 2012 -2023 (Hot and Cold side)
Trans Thai Malaysia GPP BLCAR
MCOT Vessel Control Scheme Change
Erb West,Semarang BLCA
Ammonia PCFK OJT and Step Test
Butanol DCS Migration
PCLDPE APC Assessment and APC Revamp
PCMSB APC Implementation
ABF PEMS Implementation

Spend most of time doing project at Ethylene Plant.

Competency Level

Previous Score:	0	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Can discuss the purpose of PIMS & their basic functions as well as the types of the PIMS available within PETRONAS.
Demonstrated how to display and extract the data. Explained data compression & exception. Explained the PIMS architecture and what/how monitoring dashboard was developed using PIMS data. Shared his monitoring spreadsheet to monitor inferential & process data during step test or for performance monitoring.
Explained system management tools within PIMS (e.g. PI-SMT). Shared his experience in configuring new tags in IP21 for CO inferential.

K1	Scouting Study / Functional Design / Benefits Analysis*
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Assessee Rating Description:*

Has implemented APC for tasks below:

PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section
PCLDPE Benefit Calculation
PCMSB Plant 1 and Plant 2 APC Implementation

Competency Level

Previous Score:	2	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Explained APC implementation at the ethane crackers. Explained briefly the APC implementation step. Explained furnace APC revamp due to moving some MVs to the baselayer level to simplify the structure. Elaborated that he executed the APC readiness and benefit study. Detailed out the APC structure during feas study and how the benefit is computed taking into acct reality check to see if its achievable; subsequently how the payback period is calc.
For FDS, explained the content of the FDS and how the step test plan is generated from this doc.
Amin meets Skill level.

K2	Plant Test and Modelling*
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Assessee Rating Description:*	Has implemented APC for tasks below: PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section PCLDPE Benefit Calculation PCMSB Plant 1 and Plant 2 APC Implementation
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Competency Level

Previous Score:	3	Self:	4	Assessment Target:	4	Assessor:	4	Gap:	0
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Assessor Remarks:*	Basis of up to Skill level has been established during the 1st assessment. He is also an Aspen DMC3 certified user, undergoing an exam under Aspen. Explained the importance of step test in obtaining good models and avoiding correlated moves. Explained different methods of step test and have utilised smart step for PCESB Furnace ctrlr revamp. He also led the development of the smart step guidelines document. Explained modeling process in Aspen DMC+. Explained the different concept of modelling (parametric & non-parametric) and shared his experience in comparing the diff functions (FIR, subspace, F2P & parametric). Explained the modelling steps and what guide Aspen provided to indicate model goodness. Able to explain impact of noise to modelling and later, to control/optimisation. Explained how doing DCS filter on APC handles and what needs to be done on the APC models/ctrlr. Able to explain impact of non-linearity & what he has done in furnace ctrlr to transform the damper MV and CO transformation. Overcome issue with noisy reading by filtering the data and do the calc tag in the model. Amin meets Advanced level.
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K3	Controller Build / Simulation*
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Assessee Rating Description:*

Has implemented APC for tasks below:

PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section
PCLDPE Benefit Calculation
PCMSB Plant 1 and Plant 2 APC Implementation

Competency Level

Previous Score:	2	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Briefly explained how controller was built and how he built the controller for the furnace in Aspen. He also explained the advantage of building a sub-ctrlr. Able to explain tuning of the controller and how SmartTune helps in providing the overall optimisation for the ctrlr. Explained importance of simulation. Demonstrated COT simulation where tuning changes were done to confirm with panelmen on how the ctrlr dynamics. Able to explain how to simulate disturbance that impacts the CV and how the ctrlr responds based on the tuning put in. Demonstrated what he has done to the MV logics where the issue is to avoid windup for the burner MV.
Amin meets Skill level.

K4	Implementation and Commissioning
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Assessee Rating Description:*	Has implemented APC for tasks below: PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section PCLDPE Benefit Calculation PCMSB Plant 1 and Plant 2 APC Implementation
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Competency Level

Previous Score:	2	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*	Provided overview for commissioning for APC. Explained the difference btwn ctrlr installation in PACE & DMC3 where its a simpler deployment for DMC3 since DMC3 can do the test connection. Explained what was done during FAT & SAT. Elaborated what was done during commissioning. Can explain the sequence of MV/CV commissioning, e.g. preference to turn on the easiest/least complex 1st. Also highlighted issues during commissioning. Shared HSE activity that has to be carried out prior to commissioning. Can describe what areas are covered in RA. Amin meets Skill level.
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K5	APC Integration with DCS
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Assessee Rating Description:*

Has implemented APC for tasks below:

PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section
PCLDPE Benefit Calculation
PCMSB Plant 1 and Plant 2 APC Implementation

Competency Level

Previous Score:	0	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Shared his experiences in this inventory; he has commissioned APC in PCESB, PCML, PCLDPE. Explained the h/w & s/w reqt for APC, Purdue Model, min cybersec reqt. Explained issues with antivirus update affecting apc applications and how this is done. Detailed out issues using OPC classic and what can be done or has been done to overcome this. explained how to config data exchange btwn OPC client & server. Explained what he found for Foxboro - Aspen APC ctrlr which used Foxbridge comm with OPC. The Foxbridge is holding the APC ver u/grade because it is not ready for v14 and due to this, OGD plans to incl the u/grade by using CIMIO. Also explained what needs to be monitored during the integration work to check the data flow. He also explained the configurations steps during the APC installation - logics in DCS, buffer tags creation, what needs to check to see if APC is pushing to right direction before running the APC to optimisation mode. Explained the logics for APC fail safe config like watchdog, fallback. Explained configuration steps required in installing inferential.
Amin meets Skill level.

K6	APC Performance Monitoring*
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Assessee Rating Description:*

Has implemented APC for tasks below:

PCESB C402 APC Revamp / COLD APC Revamp/ Hot Section
PCLDPE Benefit Calculation
PCMSB Plant 1 and Plant 2 APC Implementation

Competency Level

Previous Score:	0	Self:	3	Assessment Target:	3	Assessor:	3	Gap:	0
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Assessor Remarks:*

Able to explain the importance of performance monitoring and need for web-based support for operations which is governed by cybersec reqts. explained diff btwn OT & IT and how it impacts perf monitoring for APC. Detailed out what needs to be monitored for APC and how it is done (aspenwatch, excel or other tools). Explained importance of data encryption & cybersec reqts. Explained the levels of perf monitoring to ensure APC is pushing to constraints - uptime of ctrlr and the CV/MV; constraint monitoring. Can explain concept of defense in depth.
Amin meets Skill level.

K7	Regression (Statistical) Property Inferential (RPI)*
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Assessee Rating Description:*	Has developed inferential and close loop activity as per activity below: ABF PEMS Implementation PCLDPE APC Implementation PCESB C2 Soft sensor – close loop PCMSB EtOH in MeOH CO Inferential by using linear and non linear method
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Competency Level

Previous Score:	0	Self:	4	Assessment Target:	4	Assessor:	4	Gap:	0
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Assessor Remarks:*

Has carried out 7 closed loop application in PCEPE, PCMSB & PCLDPE which is utilised in APC. Have done PEMS applications. Explained the need for inferential and the issues with inferential/lab sampling. Explain advantage of inferential vs online analyser and lab samples. Explain the steps taken in building an RPI:

1. Data Collection - how to determine whether data set is good enough, volume of data required to build a good working RPI.
2. Data conditioning - explained how data can be randomised, how to choose data segment for training & validation
3. Modelling - explained steps how modelling is done and described the statistics (p-value, VIF, R2), explained collinearity, model acceptance

Explained how bias updating is done, and identify whether it is sufficient to update the bias or to revamp the inferential model by monitoring the model error. Able to explain issues with bias updating either with lab or online analyser.

Went through emicon CO inferential development which uses BDN method. Explained for the CO inferential, how to determine goodness of fit, how bias updating was done. Can explain what is PLS.

Explained PCEPE C2 loss inferential to provide a different method of modelling (using PLS, previously uses BDN). Explained the limitations of the installed RPI, LIMS integration (the how-to and the issues). Able to explain the need for advanced modelling method using neural net.

Amin meets Advanced level.

K8	First Principle Inference (FPI)
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Assessee Rating Description:*

Has attended Regression and First Principle training

Competency Level

Previous Score:	0	Self:	2	Assessment Target:	2	Assessor:	0	Gap:	2
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Assessor Remarks:*

not assessed

K9	Multi-Unit Optimisation
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Assessee Rating Description:*

Has involved and improved CLP in PCEPE

Competency Level

Previous Score:	0	Self:	2	Assessment Target:	1	Assessor:	1	Gap:	0
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Assessor Remarks:*

Explained how multi-unit optimisation helps to achieve enhanced stabilisation and what are the model types within this. Also able to discuss how supply chain impacts optimisation.
Amin meets Awareness level.

Assessment Summary

Overall Remarks

Amin did well during this assessment. He provided evidences of his work that demonstrated his capability in meeting Staff level.

Short Term potential (1-2 years)

Staff

Assessor(s)

Andriny binti Rusman
Nazatul Sheema binti Che Din

Ultimate Potential

Custodian

Supervisor

Normaya binti Shabudin

Level Achieved

Staff

Technology Class (Advanced Process Control)	Senior Executive	Staff	Principal	Custodian
Base	83%	83%	67%	17%
Key	89%	89%	33%	11%
Pacing	100%	100%	0%	0%
Emerging	100%	100%	0%	0%
Compulsory TIs	100% (7/7 met)	100% (7/7 met)	71% (5/7 met)	29% (2/7 met)

Legend: m: Meet Target m*: Exceed Target (n): Gap

Tech Class	Technology Inventory (Advanced Process Control)	Compulsory TI	Assessment Score	Senior Executive			Staff			Principal			Custodian		
B1	Process Technology*	Yes	3	2	m*	83%	3	m	83%	3	m	67%	3	m	17%
B2	Basic and Intermediate Regulatory Control*	Yes	4	3	m*		4	m		4	m		5	(1)	
B3	PID Tuning		4	3	m*		4	m		4	m		5	(1)	
B4	DCS Configuration		3	2	m*		3	m		3	m		4	(1)	
B5	Complex and Advanced Regulatory Control		0	2	(2)		3	(3)		3	(3)		4	(4)	
B6	Plant Information Management System		3	3	m		3	m		4	(1)		4	(1)	
K1	Scouting Study / Functional Design / Benefits Analysis*	Yes	3	3	m	89%	3	m	89%	4	(1)	33%	5	(2)	11%
K2	Plant Test and Modelling*	Yes	4	3	m*		4	m		4	m		5	(1)	
K3	Controller Build / Simulation*	Yes	3	3	m		3	m		4	(1)		5	(2)	
K4	Implementation and Commissioning		3	3	m		3	m		4	(1)		5	(2)	
K5	APC Integration with DCS		3	3	m		3	m		4	(1)		4	(1)	
K6	APC Performance Monitoring*	Yes	3	2	m*		3	m		3	m		4	(1)	
K7	Regression (Statistical) Property Inferential (RPI)*	Yes	4	3	m*		4	m		4	m		4	m	
K8	First Principle Inferentials (FPI)		0	1	(1)		2	(2)		3	(3)		3	(3)	
K9	Multi-Unit Optimisation		1	1	m		1	m		2	(1)		4	(3)	
P1	Expert System		1	1	m	100%	1	m	100%	2	(1)	0%	2	(1)	0%
P2	Non-Linear MPC		1	1	m		1	m		2	(1)		2	(1)	
E1	Parallel Coordinates		1	1	m	100%	1	m	100%	2	(1)	0%	2	(1)	0%
E2	Advanced Analytics for Optimization		1	1	m		1	m		2	(1)		2	(1)	

Development Plan 1

Immediate (1-2 years)

Amin is ready to become a Staff TP

Long term (More than 2 years)

He can grow to become a Custodian engineer if he wishes to pursue the technical career ladder

Development Plan 2 (Gap Analysis)

Expectation Level Staff

Legend

SL: Self Learning **CM:** Coaching and Mentoring **FT:** Formal / Classroom Training / Further Studies **SC:** Seminar / Conference **JR:** Job Rotation

Tech Class	Technical Inventory	Level Achieved	Expectation	Gap	Urgency	Development Activities	Knowledge / Skill to be acquired	Duration (months)
B1	Process Technology*	3	3	0		SL CM FT SC Jr Project		0
B2	Basic and Intermediate Regulatory Control*	4	4	0		SL CM FT SC Jr Project		0
B3	PID Tuning	4	4	0		SL CM FT SC Jr Project		0

B4	DCS Configuration	3	3	0		☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project		0
B5	Complex and Advanced Regulatory Control	0	3	3	Low	☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☒ Project	To be involved in complex loop modification project.	12
B6	Plant Information Management System	3	3	0		☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project		0
K1	Scouting Study / Functional Design / Benefits Analysis*	3	3	0		☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project		0
K2	Plant Test and Modelling*	4	4	0		☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project		0

K3	Controller Build / Simulation*	3	3	0		SL CM FT SC Jr Project		0
K4	Implementation and Commissioning	3	3	0		SL CM FT SC Jr Project		0
K5	APC Integration with DCS	3	3	0		SL CM FT SC Jr Project		0
K6	APC Performance Monitoring*	3	3	0		SL CM FT SC Jr Project		0
K7	Regression (Statistical) Property Inferential (RPI)*	4	4	0		SL CM FT SC Jr Project		0

K8	First Principle Inference (FPI)	0	2	2	Low	☒ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project	Learning from web, or MRCSB colleagues to know more about FPI.	9
K9	Multi-Unit Optimisation	1	1	0		☐ SL ☐ CM ☐ FT ☐ SC ☐ Jr ☐ Project		0
P1	Expert System	1	1	0	Low	☒ SL ☒ CM ☐ FT ☐ SC ☐ Jr ☐ Project	NA. Gap closed.	6
P2	Non-Linear MPC	1	1	0	Low	☒ SL ☒ CM ☐ FT ☐ SC ☐ Jr ☐ Project	NA. Gap closed.	6
E1	Parallel Coordinates	1	1	0	Low	☒ SL ☒ CM ☐ FT ☐ SC ☐ Jr ☐ Project	NA. Gap closed.	6

E2	Advanced Analytics for Optimization	1	1	0	Low	☒ SL ☒ CM ☐ FT ☐ SC ☐ Jr ☐ Project	NA. Gap closed.	6
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